

OC Core 1.3.2 Expert Technical Spine

Formal Navigation for Reviewers of the Release Candidate Package

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Technical Role

This spine gives a compact expert route through OC Core 1.3.2. It is meant for reviewers who want to inspect the formal and evidential architecture before reading the entire master monograph.

Core Grammar

OC Core represents a continuum through identity-bearing conditions, admissible realizations, thresholds, embeddings, and history-sensitive classification. The release-relevant grammar distinguishes:

- live continuation under preserved admissible realization;
- death or collapse when admissible realization is empty;
- residue after death without live identity;
- rebirth when later live structure is grounded in residue without being identical to the original continuum.

Claim-Support Classes

Class	Release meaning
FORMALLY_PROVED	The release contains or references a proof route adequate for the claim scope.
THEOREM_NATIVE_HELD_OUTSIDE_VALIDATED	The VALIDATED is retained only where bounded proof support exists.
OPERATIONALLY_SUPPORTED_WITHIN_BOUNDS	Empirical support is present, but the claim remains bounded by protocol and falsifier.
SIMULATION_ILLUSTRATION_ONLY	Simulation clarifies behavior without serving as standalone proof.
PUBLIC_ROUTE_DISCOVERY_ONLY	Only route is exposed for audit, not used as strong support.
FRONTIER_WORK	Future research surface; not a canonical Core promotion.

Release Technical Invariants

- No canonical claim is promoted by a support packet alone.
- No prediction claim is accepted without replay protocol, code, fixture or data, output hash, tolerance envelope, and falsifier.
- No external criticism response is treated as closure unless release-visible evidence, proof route, replay, or owner-gate terminal state exists.
- No private reviewer-source material or private model transcript is included in the public package.
- No publication action is lawful while the no-send lock remains true.

Primary Artifact Map

Artifact	Expert use
Master monograph	Full formal narrative, definitions, proof route, domain appendices, and reviewer navigation.
Journal core	Compact source-audited theory kernel.
Methods companion	Build, replay, source-audit, and release-governance route.
Critique map	Objection families, response surfaces, and blocker ledger reading order.
Release dossier	Artifact inventory, gates, no-send lock, and owner action route.
Science blocker closure ledger	Row-level terminality of the release blockers.

Owner Review Boundary

OC Core 1.3.2 can be reviewed as a coherent release package. It still cannot publish itself. Owner approval, publication channel decision, and final artifact freeze confirmation are separate lawful actions.

Reviewer Routes

Reviewer	Technical route
Owner	No-send lock, owner packet, package freeze hash, release dossier, and publication risk memo.
Mathematical reviewer	Formal definitions, theorem roadmap, proof machinery appendix, closure ledger, and support map.
Domain reviewer	K/domain projection matrix, replay manifests, DRT strict salvage, and domain packet audit.
Reproducibility reviewer	Manifest, checksums, ZIP integrity, data/simulation manifests, and PDF text-quality scan.
Publication reviewer	LRGEF, Parfitian Cerberus, citation metadata, release notes, and postflight checklist.

Proven, Replayed, Future

Class	Technical interpretation
Proven or evidence-closed	Release support is anchored in definitions, theorem-fate rows, claim-ledger support, or the terminal closure ledger.
Replayed	Release support is anchored in deterministic package artifacts, data or fixture manifests, output hashes, and tolerance/falsifier records.
Future work	The item is deliberately outside 1.3.2 authority and may only become stronger through a later release gate.

Support-Map Discipline

The expert reader should use the prediction and promotion support map as a guardrail. It does not make weak statements strong. It binds risky words to their allowed support route: proof, replay, claim ledger, structural-only boundary, or future-work route. Any statement that cannot be bound must be rewritten before release.

Admitted 1.4 Formal Layer in 1.3.2

The release imports the parts of the OC 1.4 forward layer that are already formal, benchmarked, and public-safe. They are not treated as a separate authority; they strengthen the 1.3.2 evidence contract.

Observation architecture. An observation architecture is represented as

$$A = (\Omega, \mathcal{E}, \mathcal{O}, \mathcal{R}, \Theta, \Phi),$$

where Ω is the event space, $\mathcal{E}$ the admissible evidence surface, $\mathcal{O}$ the observer or instrument family, $\mathcal{R}$ the reference class, Θ the threshold and tolerance parameters, and Φ the declared falsifier family.

Fact contract. For event E and history H_t , the release-admissible fact relation is:

$$\text{Fact}_{A,\mathcal{R}}(E, t) \iff S_{A,\mathcal{R}}(E \mid H_t) \geq \tau_{A,\mathcal{R}} \wedge \forall \varphi_i \in \Phi, \varphi_i(E, H_t) = 0.$$

This gives reviewers an explicit support/falsifier form: a fact is not merely asserted; it must cross the declared stabilization threshold and survive the active falsifier set.

Objectivity metric. For observer or instrument posteriors $P_i(E)$ in a reference class $\mathcal{R}$:

$$C_{\mathcal{R}}(E) = 1 - \frac{2}{n(n-1)} \sum_{i < j} JS(P_i(E), P_j(E)),$$

with JS the Jensen–Shannon divergence. In the deterministic 1.4-to-1.3.2 benchmark fixture, the multi-observer objectivity task records $C_{\mathcal{R}} = 0.999895$ and mean $JS = 0.000105$.

Global Verdict-Invariant Minimality Theorem

Theorem. For the class of OC-compatible verdict-preserving representations, each primitive distinction in the eight-row witness matrix below is globally necessary up to definitional equivalence: any representation that removes the distinction must re-encode functionally equivalent information or it cannot preserve the required OC verdicts.

Proof. Let π_p be one primitive distinction and let (H_a, H_b) be its separating witness pair. By construction, H_a and H_b agree on all primitive distinctions except π_p , while the OC verdict map requires $V(H_a) \neq V(H_b)$. Suppose a representation R removes π_p and carries no functionally equivalent invariant. Then $R(H_a) = R(H_b)$. Any downstream verdict decoder D therefore gives $D(R(H_a)) = D(R(H_b))$, contradicting $V(H_a) \neq V(H_b)$. Hence every verdict-preserving OC representation must preserve π_p up to definitional equivalence. Applying the argument to the eight independent witness axes proves the global lower bound for the OC verdict class.

Axis	Primitive	Witness collapse if removed
MIN-AXIS-01	identity carrier	Live continuation and unrelated replacement with matching observable state become indistinguishable.
MIN-AXIS-02	admissible realization	Death/collapse cannot be separated from constrained survival.
MIN-AXIS-03	collapse threshold	Below-threshold perturbation and transition-causing collapse lose a falsifiable boundary.
MIN-AXIS-04	residue relation	Residue-supported rebirth and unrelated birth become observationally identical.
MIN-AXIS-05	rebirth predicate	Rebirth and persistence collapse into one label.
MIN-AXIS-06	K-level boundary	Isomorphic operators in different domains lose typed falsifier discipline.
MIN-AXIS-07	support class	Toy support and independently reproduced support become public-worded at the same strength.
MIN-AXIS-08	falsifier	Pass/fail status for prediction-like language is not definable.

Deterministic Benchmark Suite

The admitted numerical layer is deterministic and reproducible. It consists of ten fixed-seed tasks: toy sharpening, double-slit toy, Stern–Gerlach toy, POVM instrument, continuous measurement, decoherence pointer basis, entanglement no-signalling, multi-observer objectivity, architecture perturbation, and cognitive order effect. The current run records ten runnable tasks, nine accepted baselines, zero failures, and output hash

9381f1b2df7b14792cfc307d647aa57938583f7e94c56bd5acc345b76bb01262.

The no-signalling maximum in the entanglement/instrument fixtures is 0.0. These numbers support the operational architecture and benchmark discipline; they do not by themselves assert replacement physics.

Technical Acceptance Criteria

- six primary PDFs have current identity, adequate page count, and clean extracted text;

- all eighty-two closure rows carry row-level support class, source anchor, evidence refs, manuscript refs, release refs, and terminal rationale;
- the ZIP contains no unclassified legacy primary PDFs or nonpublic template payloads;
- every strong prediction or promoted-claim occurrence is represented in the support map;
- public release remains impossible until owner/channel actions are explicitly unlocked.

Technical Appendix A: Formal Review Table

Formal surface	Question to ask
Definitions	Are identity, admissible realization, collapse, residue, and rebirth defined before use?
Theorem route	Is every theorem-native phrase bound to a theorem-fate or proof-support row?
K-level system	Does each K-level projection state match the domain and inclusion ledgers?
Claim ledger	Does each promoted claim carry an allowed support class and evidence reference?
Release machine	Does the gate suite pass without enabling publication?

Technical Appendix B: Evidence Review Table

Evidence type	Sufficiency standard
Proof route	Source-audited formal statement, theorem-fate classification, and clear release scope.
Replay	Fixed inputs, executable or documented route, output hash, tolerance, and falsifier.
Source audit	Predecessor/source tradition named with bounded influence on OC claims.
Strict salvage	Workflow and reproducibility evidence without automatic Core promotion.
Owner gate	Explicit no-send terminal state and no outbound action.

Technical Appendix C: Decision Boundary

The release is technically strong when its evidence topology is inspectable and every release-admissible strong claim has a proof, replay, or explicit source route. Unsupported strong phrases cannot become public authority. That is the purpose of the claim ledger, support map, closure ledger, platinum science audit, and release gates working together.